

Internet of Things and Vehicular Networks (TP-546)

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YOUR PHONE TOLD YOUR SMARTWATCH,
WHICH TOLD YOUR SMART HOME
AUTOMATION SYSTEM, WHICH TOLD ME
THAT YOU OWE YOUR WIFE AN
APOLOGY.

TWO DOZEN RED FLOWERS ARE JUST
R\$120.00 FOR A LIMITED TIME.



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FISH
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IoT Security



- Kaspersky detects more than 100 million attacks on smart devices in the first half of 2019;
- This number is seven times higher than the number found in the first half of 2018, when only about 12 million attacks originating from 69,000 IP addresses were detected. Taking advantage of the weak security of IoT products, cybercriminals are intensifying their attempts to create and monetize IoT botnets;
- Researchers were able to locate the regions that most frequently became sources of infection in H1 2019. These are China, with 30% of all attacks occurring in this country, Brazil with 19%, followed by Egypt (12%).
- In the first half of 2018, the situation was different, with Brazil leading at 28%, China second at 14%, and Japan next at 11%..

A USB flash drive can compromise a company's security.

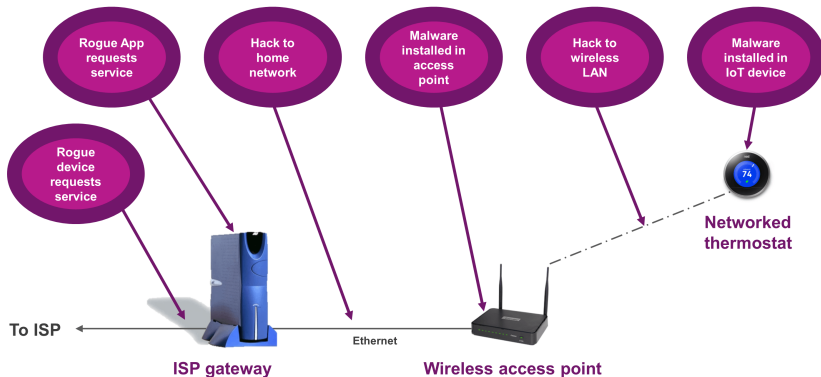


Characteristics of IoT devices:

- Low power consumption;
- Low computing power;
- Low cost;
- Plug and play is desirable.

Difficulty implementing policies related to security and privacy

Security attacks



- IoT devices for children are a common target for hackers;
- Always-listening devices;
- Location.

Devices for children:



Attackers spoof calls
track current location

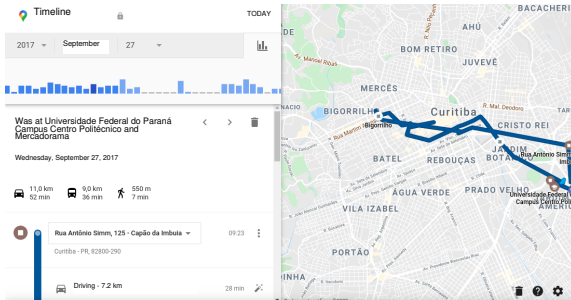


Attackers have access to image

Always-listening devices:



Location:



Desired security properties:

- Confidentiality: The property that ensures information is not disclosed or made available to any unauthorized entity.
- Integrity: The property of safeguarding the accuracy and integrity of assets in an IoT system.
- Availability: Refers to the property that ensures an IoT device or system is accessible and usable on demand by authorized entities.
- Authenticity: The assurance that an information transaction comes from the source it claims to come from. Device authenticity can be verified through authentication, which involves proof of identity.

- Authorization: A property that determines whether a user or device has the rights / privileges to access a resource or issue commands.
- Accountability: The property that ensures each action can be attributed to a single user or device.
- Reliability: Refers to the property that ensures the consistent intended behavior of an IoT system.
- Privacy: In the context of IoT, privacy refers to the user's control over the disclosure of their data.

Mechanisms to ensure security:

- Authentication: The security service that verifies the identity of a user or device.
- Access control: In the context of IoT, access control refers to the security mechanism that limits the actions or operations of a legitimate user or device in an IoT system, as well as setting a limit for programs executed on behalf of the legitimate user.
- Encryption: The process of translating or encoding information into a secret code so that only authorized entities can decode it.
- Secure boot: This security feature allows a device to verify each piece of software using a digital signature, including the operating system, when the device is first turned on or whenever it restarts.

Mechanisms to ensure security:

- Secure updates: Ensure that smart devices authenticate security patches from operators using digital signatures so that patches cannot be intercepted, extracted, and modified.
- Backup: The copying and archiving of valuable data for the purpose of restoring the original in the event of data loss.
- Device tamper detection (DTD);

Internet of Things Security — Ken Munro — TEDxDornbirn

<https://youtu.be/pGtnC1jKpMg>

Spies and Dolls: The Future of IoT Security — Maire O'Neill —
TEDxQueensUniversityBelfast

<https://youtu.be/E0FJ5K16pYE>

Types of attacks:

- Physical attacks
- Cryptographic attacks
- Botnets
- DoS (denial of service)
- Firmware hijacking
- Man-in-the-Middle
- Ransomware
- Eavesdropping
- Privilege escalation
- Brute-force attack to discover passwords

Physical attacks

- Physical attacks occur when IoT devices can be physically accessed by anyone.
- With most cybersecurity attacks occurring from within a company, it is essential that your IoT devices are in a protected area, which is often not an option.
- Many physical cybersecurity attacks begin with an attacker inserting a USB drive to spread malicious code, so it is more important than ever to add AI-based security measures to ensure your devices and data are protected.

Cryptographic attacks

- When an IoT device is not encrypted, an intruder can observe the data and capture it for later use.
- Furthermore, “once encryption keys are unlocked, cyberattackers can install their own algorithms and take control of your system”.
- For these reasons, encryption is essential in the IoT environment as part of your cybersecurity efforts.

- Botnets

Consider the Mirai botnet attack, which turned networked IoT devices into remotely controlled bots that can be used as part of a botnet.

- Botnets have the ability to use smart, connected devices to transfer confidential and private corporate data, which can be sold on the dark web, or to disable a device.
- Mirai remains a problem today, with millions of IoT devices affected.

What is Mirai and How do You Protect Yourself Against it?

<https://youtu.be/KKSyB476n9o>

DoS (denial of service)

- A DoS attack occurs when a service, such as a website, becomes unavailable.
- A large number of systems attack a target through botnets, requesting a service at the same time.
- Although attackers in this case typically do not aim to capture data, they seriously affect businesses if services become unavailable.

- A new high-severity vulnerability affects Philips Hue Smart Light Bulbs and can be exploited over the air from more than 100 meters away to enter a targeted WiFi network.
- The underlying high-severity vulnerability, tracked as CVE-2020-6007, lies in the way Philips implemented the Zigbee communication protocol in its smart bulb.



- The hacker controls the bulb's color or brightness to make users think the bulb is malfunctioning.
- The bulb appears as "Unreachable" in the user's control app, so they will try to "reset it".
- The only way to reset the bulb is to delete it from the app and then instruct the control bridge to rediscover it. The bridge discovers the compromised bulb, and the user adds it back to the network.

- The hacker-controlled bulb with updated firmware then uses vulnerabilities in the ZigBee protocol to trigger a buffer overflow by sending a large amount of data to the bridge. This data also allows the hacker to install malware on the bridge, which in turn is connected to the target company or home network.
- The malware connects back to the hacker and, using a known exploit, can infiltrate the target IP network to spread ransomware or spyware.

Hacking Smart Light Bulbs — Latest Research from Check Point
https://youtu.be/4CWU0DA__bY

Zigbee War Flying

<https://youtu.be/Ed10jAuRARU>

- LIFX Mini smart bulbs reveal home Wi-Fi passwords and basically have no security.
- The bulbs store Wi-Fi access passwords in plaintext, have no security settings, and hard-code a private encryption key directly into the firmware;
- Credentials are passed from one networked bulb to another in a mesh network powered by 6LoWPAN.

Firmware hijacking

- If you are not keeping up with IoT firmware updates, you risk a cybersecurity attack.
- Make sure you check that your updates come from the expected source. Otherwise, an attacker can hijack the device and download malicious software.
- Most hardware manufacturers do not cryptographically sign embedded firmware.

Man-in-the-Middle

- A man-in-the-middle attack occurs when a hacker breaks the communication between two separate systems.
- By secretly intercepting communications between two parties, this type of attack makes the recipient think they are receiving a legitimate message. In other words, the man in the middle starts communicating with both parties, hence the name. It may look like an email from your bank asking you to log in to perform a task.
- Now the attackers' fake website collects your credentials, allowing the attacker to cause further damage.

- In 2015, a security flaw was discovered that allowed hackers to access a user's Google credentials through a Samsung smart refrigerator. The refrigerator had an integration with Google Calendar;
- The refrigerator's software provided SSL (Secure Sockets Layer) encryption to establish an encrypted link, but in reality it completely failed to validate SSL certificates.



Ransomware

- Ransomware is a type of malware that blocks access to files by encrypting them.
- Attackers then sell the decryption key so that your files can be accessed again.
- Naturally, this type of attack can disrupt day-to-day business, and the encryption key usually comes at a high price.

When coffee makers are demanding a ransom, you know IoT is screwed

<https://arstechnica.com/information-technology/2020/09/how-a-hacker-turned-a-250-coffee-maker-into-ransom-machine/>

What a hacked coffee machine looks like.

<https://youtu.be/bJrIh94RSiI>

<https://www.smarter.am/coffee>

Eavesdropping

- In this type of attack, a hacker intercepts network traffic to steal confidential information through a weakened connection between an IoT device and a server.
- Eavesdropping is usually carried out by listening to digital or analog voice communications or by intercepting detected data.
- Again, in this case, the attacker leaves with confidential corporate data.

Privilege escalation

- Hackers look for bugs and weaknesses in an IoT device to gain access to resources that are normally protected by an application or user profile.
- In this type of attack, the hacker seeks to use their newly acquired privileges to deploy malware or steal confidential data.

Brute-force attack

- In this scenario, hackers submit many passwords or passphrases in the hope of guessing the correct one, giving them access to your IoT devices.
- Or they use software to generate a large number of consecutive guesses.
- Now that the attacker has access to your device, they can install malware or steal business-critical data.

Brute-force attack

- Kaji Malware;
- Chinese origin;
- Go programming language;
- Under development;
- Does not compromise the IoT device;
- Brute-force attack to carry out DDoS attacks.

Strava

How a Fitness App's Heat Map Uncovers Military Bases — NYT

<https://youtu.be/0IB8p-YpXwA>

WEBINAR recording - IoT Security: the Key Ingredients for Success

<https://youtu.be/bcVzt141JN8>

Assignment

- Research security issues in IoT networks. Choose one type of security issue. Show how a specific attack can be carried out and which measures can be used to address the problem.